Arthur Santana, Ph.D.

Computational Linguist | NLP & LLM Systems | AI Evaluation & Quality
Prompt Engineering

arthurpereirasantana@gmail.com https://arthurpsantana.github.io/
+55 (11) 94621-6111 https://www.linkedin.com/in/arthursantana

Bio

Computational linguist with 8+ years of experience building NLP and LLM systems, multilingual evaluation pipelines, semantic annotation frameworks, and AI quality processes for large-scale language technologies.

Skills

- Linguistics
Phonetics, Phonology, Syntax, Morphology, Portuguese

- Computational Linguistics
Language Data Analysis, Experiment Design, Language Modeling, Semantic Annotation, Taxonomy Design Prompt Engineering

- **NLP**
NLU, NLG, ASR, TTS

- **Technical**
Python, Version Control (Git), R, Praat, SQL

Education

- Ph.D in Linguistics
University of São Paulo
(2015-2019)

- Visiting Ph.D. Scholar
University of Southern California
(2017-2018)

- M.A in Linguistics
University of São Paulo
(2013-2015)

- B.A in Letters
Federal University of Maranhão
(2008-2013)

Work Experience

☐ **AI Quality Assurance Specialist** at Decagon (via Mercor)
(Nov/2025 - Current)

- Conduct quality assurance evaluations of linguistic data and AI model outputs, ensuring compliance with project guidelines.
- Evaluate AI agent performance using structured rubrics and predefined quality standards across conversational and NLP tasks
- Identify recurring error patterns providing actionable, structured feedback to improve model performance
- Collaborate with cross-functional stakeholders to clarify requirements, resolve ambiguities, and refine QA and evaluation procedures
- Maintain detailed documentation of evaluation criteria, decisions, and edge cases to support auditability and consistency.

☐ **Applied NLP Engineer** at Center for Curriculum Redesign
(Oct/2024 - Current)

- Design and deploy multi-LLM evaluation pipelines to improve reliability, factual accuracy, and consistency of AI-generated educational content;
- Deployed structured prompt workflows and evaluation pipelines that coordinated multiple LLMs to validate, refine, and improve AI-generated outputs through iterative consensus and rubric-based assessment.
- Expanded and expanded hierarchical competency models and label structures used for automated evaluation, enabling scalable and auditable assessment pipelines.
- Built automated workflows for multi-model comparison, including sequential refinement and consensus-based architectures for robust output validation;
- Developed evaluation frameworks using hierarchical competency taxonomies and adaptable scoring rubrics for AI-generated educational content.
- Implemented hallucination detection and correction systems, leveraging structured taxonomies and evaluator-based validation strategies;
- Engineer prompting strategies and structured outputs (JSON) to ensure deterministic, auditable, and scalable model behavior across tasks;
- Adapted prompt templates, evaluation criteria, and structured label systems to ensure consistent behavior across different LLMs and multilingual educational datasets.
- Integrate pipelines with GitHub Actions and external data systems (e.g., Airtable), enabling automated runs, experiment tracking, and longitudinal benchmarking.

☐ **Quality Controller** at Telus Digital
(Jun/2025 - Nov/2025)

- Lead quality assurance for language data projects, ensuring accuracy, consistency, and alignment with client specifications;
- Collaborate with project managers and subject matter experts to refine task guidelines, documentation and resolve quality issues;
- Conduct audits and performance reviews of annotators and reviewers;

Languages

- Brazilian Portuguese
Native

- English
Fluent/Bilingual

- Spanish
Intermediate Level

Grants

- Ph.D. Research
Conselho Nacional de Desenvolvimento Científico e Tecnológico.
Process n. 140139/2015-0
(2015-2019)

- International Visiting Scholar Grant
Conselho Nacional de Desenvolvimento Científico e Tecnológico.
(2017-2018)

- M.A. Research
Conselho Nacional de Desenvolvimento Científico e Tecnológico.
Process n. 130194/2013-1
(2013-2015)

- Analyze error patterns and generate insights to improve data quality and team efficiency;
- Support the onboarding and training of new team members, ensuring alignment with quality standards;
- Create, update, and edit annotation and QA guidelines to reflect evolving project needs and improve clarity for annotators
- Use quality tracking tools and dashboards to report on KPIs, monitor SLA compliance, and drive continuous improvement

☐ **Senior Linguist** at Google Brazil (via Telus Digital)
(oct/2023 - Jun/2025)

- Support the Project Manager on management tasks;
- Perform review of work authored by Linguists;
- Elaborate and conduct training activities of Linguists and new-hires;
- Localized prompts, response templates, evaluation instructions, and linguistic resources for Brazilian Portuguese, ensuring linguistic accuracy, cultural appropriateness, and consistency with global AI quality standards.
- Develop complex grammars for Natural Language Understanding;
- Create complex templates for Natural Language Generation;
- Adapted prompts, response templates, and evaluation instructions for Brazilian Portuguese, ensuring linguistic accuracy, cultural appropriateness, and consistency with global AI quality standards.
- Evaluate current system outputs;
- Annotate and review linguistic data.
- Follow and fulfill project KPIs

☐ **NLU Language Developer** at Cerence (via Maize)
(jun/2022 - aug/2024)

- Developed NLP components: write rules, build grammars/FSTs
- Expanded annotation schemas and semantic label structures to support new conversational AI capabilities while maintaining compatibility with existing language resources
- Used modeling tools to bootstrap and test new functionalities
- Designed Portuguese-specific linguistic mappings and expanded semantic label structures to improve NLU coverage and speech technology performance.
- Created annotation systems and mappings for Portuguese
- Analyzed system performance and identify areas for improvement
- Collaborated with other Engineers and annotators in data collection and development efforts
- Developed specific guidelines for Brazilian Portuguese phonetic/phonemic transcription
- Supported script design used to train speech synthesis models
- Worked on localization (en-ptBR-en)

☐ **Associate Linguist** at Google Brazil (via Telus Digital)
(sep/2018 - sep/2023)

- NLU Grammar development
- NLG development
- Developed annotation guidelines, semantic label definitions, and taxonomy documentation to support scalable NLP data creation and model training.
- Annotate and review linguistic data
- Provide linguistic analysis and support for NLU and NLG projects
- Response modeling
- Structured linguistic annotations using standardized semantic labels for disambiguation, normalization, and language understanding tasks.
- Annotating lexicon entries according to guidelines
- Deriving NLP data

☐ **Computational Linguist** at Portuguese International Institute
(jan/2022 - jul/2022)

- Supported the expansion and organization of scientific terminology taxonomies for Portuguese language resources.
- Data cleaning and wrangling
- Flexional paradigm script design

☐ **Visiting PhD Scholar** (*USC, Los Angeles, United States*)
(2017-2018)

☐ **Teacher's Assistant** (USP, São Paulo, Brazil)
(2015-2019)

☐ **English teacher** (*English for Business, São Paulo, Brazil*)
(2013-2017)

☐ **English teacher** (*Cultura Inglesa, São Luís, Brazil*)
(2011-2013)

Academic Experience

Publications

Journal Articles

2018. *The status of unstressed lax-mid vowels in northern dialect of Brazilian Portuguese*. Revista Linguíʃtica.
2018. *The role of Phonological processes to discussing vowel inventories in Brazilian Portuguese*. Acta Linguistica Academica.
2016. *Uma abordagem fonológica para as postônicas não-finais*. Domínio de Linguagem.
2012. *A síncope revisitada: uma análise com base no corpus do ALiMA*. Littera.
2011. *O livro didático do português: uma análise da oralidade*. Littera.

Book Chapters

2014. *A redução de proparoxítonas no falar maranhense*. (with José Bezerra) In: In: RAZKY, Abdelhak et. al (Orgs.). Estudos sociodialetais do português brasileiro. Editora Pontes.
2013. *A permanência da tônica em variantes proparoxítonas no falar maranhense*. In: RAMOS, Conceição et. al. (Orgs.) O português falado no Maranhão: multiplos olhares. EDUFMA.

Conference Proceedings.

2012. *O apagamento das postônicas não-finais no falar maranhense*. IV Seminário Internacional de fonologia.
2012. *A neutralização das postônicas não finais no falar maranhense: uma análise preliminar*. II Congresso Internacional de Dialetologia e Sociolinguística.
2011. *Variação de proparoxítonas: traços da identidade popular no falar maranhense*. III Seminário Linguagem e identidades: múltiplos olhares.
2010. *'Risca o fósqui que a lâmpida apagô': um estudo geosociolinguístico das proparoxítonas no falar maranhense*. I Congresso Internacional de Dialetologia e Sociolinguística.
2010. *Apócope em proparoxítonos: um estudo preliminar*. I Congresso Internacional de Dialetologia e Sociolinguística.

Conference Presentations

2018 *Is there reduction via laxing in northern dialects of Brazilian Portuguese?* Generative Linguistics in the Old World - GLOW 41. (Budapest, Hungary. April, 2018)
2018 *The status of unstressed lax mid vowels in the northern dialect of Brazilian Portuguese*. Experimental Portuguese Linguistics Workshop (Toronto, Canada. April, 2018)
2018 *Syncope and the canonical foot in Brazilian Portuguese*. 7th Hispanic and Luso-Brazilian Linguistics Conference (Tempe, Arizona. March, 2018)

Last updated:
Apr, 2026

2016. *The role of phonological processes to discussing vowel inventories in Brazilian Portuguese*. Old World Conference in Phonology - OCP 13 (Budapest, Hungary. January, 2016)
2015. *Uma abordagem fonológica para as postônicas não-finais*. II Encontro intermediário do GT de Fonética e Fonologia da ANPOLL. (Uberlândia, Brazil. September, 2015)
2015. *Neutralization of non-final postonic mid vowels in Brazilian Portuguese: assimilation or feature deletion?* Phonetics and Phonology in Europe - PAPE 2015 (Cambridge, UK. July, 2015)
2015*. Vogais medias em dialetos nordestinos: as postônicas não-finais em São Luís*. 63 Seminário do GEL. (Campinas, Brazil. June, 2015).